Harry Lu

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Education

University of Waterloo Waterloo, Ontario

Candidate for Bachelor of Computing and Financial Management : Major GPA 3.90/4.00 *2025-2030*

- Recipient of the **University of Waterloo President's Scholarship ($2,000)**
- Relevant Coursework: Object-Oriented Programming, Algorithm Design & Data Abstraction, Logic & Computation

Technical Skills

Languages: Python, C, C++, SQL, HTML, CSS, Javascript
Libraries/Frameworks: Pandas, Pytorch, OpenCV, Sci-kit Learn, Matplotlib, FastAPI, React
Developer Tools: Git, Linux
Databases: SQLite, PostgreSQL
Data Visualization: Power BI, Tableau

Projects

Fitness Form Checker (FormFirst) | *Python, OpenCV, Yolo, Numpy*

- Built a workout form analyzer using **OpenCV** and **YOLO** pose estimation that detects 5 exercises (Lat Pulldowns, Pull-Ups, Dips, Deadlifts, Squats) and provides per-frame form feedback from video input
- Auto-detected exercise and rep start/end, counted reps, and provided form feedback by calculating joint angles across **17 key points** across various frames in video inputs using YOLO keypoint detection.
- Handled edge cases involving multi-person videos or persons blocked by objects by selecting the largest **bounding box** in video frame, isolating the target user and ensuring accurate tracking.

Kitchen to Recipe | *Python, Pandas, Pytorch, FastAPI, PostgreSQL*

- Designed a website that identifies food items in fridge photos using a **PyTorch** convolutional neural network (CNN) and returns simple recipes from a PostgreSQL database.
- Aggregated 16,000+ images across 4 datasets and removed **7,000+ duplicates** using perceptual hashing. Further consolidated 87 overlapping class labels into 40 standardized food categories.

News Sentiment Analysis | *Python, FastAPI, React, SQLite*

- Built a daily news sentiment pipeline that scrapes **200+** Canadian news articles from FreeNewsAPI, determines the location of publication by analyzing headlines and news body text, and then scores headlines based on sentiment.
- Designed a weighted scoring system combining **HuggingFace** NLP model sentiment with article topic severity to produce daily "stress index" scores for articles across news categories.
- Stored article data and sentiment scores in **SQLite**, served to a **React** dashboard via **FastAPI** to visualize daily trends over time. Displayed articles as interactive dots on an interactive map of Canada.
- Automated daily scraping and storing of articles using cron jobs.

Insurance Risk Portfolio Analysis | *Semi Finalist - ASNA 2026 Case Competition* | *Python, Pandas, Matplotlib*

- Analyzed a **10,000** policyholder insurance dataset across 4 coverage types using Python and Pandas and identified key risk patterns driving $8.5M in portfolio losses. Presented information using Matplotlib graphs.
- Found high correlation between fraternity style residences and losses by dividing policyholders by residence type, distance, and region to find relationships between losses and policyholder characteristics.
- Quantified "worst-case scenario" financial exposures by calculating Value at Risk and Tail Value at Risk loss at 90th, 95th, and 99th percentiles across all coverage types, finding guest-medical coverage losses exceed $**3.87M**.
- Presented strategic recommendations for excess-of-loss reinsurance and residence-based price segmentation to a panel of industry judges, advancing to semi-finals round.

Experience

Skiing Apprentice Instructor

Uplands Golf and Ski Club *Vaughan, ON*

- Assisted Senior certified Ski instructors in teaching classes and aiding students who were falling behind/had trouble learning, teaching one-on-one classes when necessary.
- Adapted teaching style to different age groups (child, teen, adult) and learning paces, communicating updates and feedback to parents and lead instructors when necessary.